

From: Tavalire, Anne (EGLE)
Sent: 11/26/2025 9:07:58 AM
To: "Jeff Dietlin" <utilities@cadillac-mi.net>
Cc: "Kline, Aaron (EGLE)" <KlineA4@michigan.gov>; "Lowe, Claire (EGLE)" <LoweC9@michigan.gov>; "Hendershott, Abigail (EGLE)" <HENDERSHOTTA@michigan.gov>; "Peterson, Amy (EGLE)" <PETERSONA1@michigan.gov>; "Ploehn, Kelly (EGLE)" <PLOEHNK@michigan.gov>; "Jury, Michael (EGLE)" <JURYM1@michigan.gov>
Subject: October 2025 PFAS sample results
Attachments: 2510354_1 ENVReport 11 24 2025 1129.pdf, 2510354 FINAL EGLE Excel 24 Nov 25 1129.xls

Hi Jeff,
Attached are the results for the PFAS WWTP effluent sample EGLE collected on October 22, 2025. The results are in compliance with the 5 PFAS with Water Quality Values.

The sample results have been uploaded to the Cadillac WWTP MiEnviro Portal under the evaluation: E-45504 Sampling Only (Toxics). Please forward to other WWTP staff as appropriate.

Let me know if you have any questions.

Thanks,
Anne Tavalire, Regional Pretreatment Program Specialist
Water Resources Division | Emerging Pollutants Section
Michigan Department of Environment, Great Lakes, and Energy (EGLE)
Telephone number: 248-508-1102
[Follow Us](#) | [Michigan.gov/EGLE](https://michigan.gov/EGLE)

ATTACHMENT NAME:

2510354_1 ENVReport 11 24 2025 1129.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

24 November 2025

Work Order: 2510354

Grace Borst
EGLE-WRD-LANSING
525 W. Allegan, P.O. Box 30242
Lansing, MI 48909-7742
RE: CADILLAC WWTP

This is the official environmental laboratory report for testing conducted by the Michigan Department of Environment, Great Lakes, and Energy. Analyses performed by the laboratory were conducted using methods published by the U.S. Environmental Protection Agency, Standard Methods for the Examination of Water and Wastewater, ASTM, or other published or approved reference methods.

Kirby Shane
Laboratory Director

Total Charges for Reported Analyses:

\$1,000.00



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

EGLE-WRD-LANSING
525 W. Allegan, P.O. Box 30242
Lansing MI, 48909-7742

Project: CADILLAC WWTP
Location ID: MI0020257
Project Manager: Grace Borst

Reported:
11/24/2025

Analytical Report for Samples

Sample ID	Laboratory ID	Matrix	Date Sampled	Date Received	Qualifier
C	2510354-01	Water	10/22/2025	10/23/2025	
TB	2510354-02	Water	10/22/2025	10/23/2025	

Qualifiers and Definitions

IDA01 Result is estimated/suspect due to isotope dilution analog recovery in the sample outside of QC limits.
A03 Result(s) and reporting limit(s) are estimated due to low matrix spike recovery.
ND Indicates the analyte was not detected at or above the method reporting limit (RL)
RL Reporting Limit
NA Not Applicable

*****Case Narrative*****

Samples were received **10/23/2025 11:20:00AM** for client **EGLE-WRD-LANSING** as a part of project **CADILLAC WWTP**.

Temperature of samples upon receipt was recorded at 3.30° C and samples were received on ice

Samples were logged and designated as Work Order # **2510354** on **10/23/2025 3:10:00PM**.

This Report was created **11/24/2025 11:29:20AM**.

Additional Notes/Narrative (if applicable):



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Client ID: C
Lab ID: 2510354-01

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Analyst	Qualifier
Organics-Semivolatiles										
763051-92-9	11Cl-PF3OUdS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
356-02-5	3:3FTCA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
757124-72-4	4:2FTS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
914637-49-3	5:3FTCA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
27619-97-2	6:2FTS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
812-70-4	7:3FTCA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
39108-34-4	8:2FTS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
756426-58-1	9Cl-PF3ONS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
919005-14-4	ADONA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
13252-13-6	HFPO-DA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
4151-50-2	NEtFOSA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2991-50-6	NEtFOSAA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
1691-99-2	NEtFOSE	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
151772-58-6	NFDHA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
31506-32-8	NMeFOSA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2355-31-9	NMeFOSAA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
24448-09-7	NMeFOSE	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-22-4	PFBA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-73-5	PFBS	4.4	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
30334-69-1	PFBSA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
335-76-2	PFDA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
307-55-1	PFDoA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
79780-39-5	PFDoS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	A03
335-77-3	PFDS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
646-83-3	PFECHS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
113507-82-7	PFEESA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-85-9	PFHpA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-92-8	PFHpS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
307-24-4	PFHxA	6.3	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
355-46-4	PFHxS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
41997-13-1	PFHxSA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
863090-89-5	PFMBA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
377-73-1	PFMPA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-95-1	PFNA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
68259-12-1	PFNS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
335-67-1	PFOA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
1763-23-1	PFOS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
754-91-6	PFOSA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2706-90-3	PFPeA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2706-91-4	PFPeS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
422-64-0	PFPrA	16.0	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
423-41-6	PFPrS	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Client ID: C
Lab ID: 2510354-01

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Analyst	Qualifier
Organics-Semivolatiles										
376-06-7	PFTeDA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
72629-94-8	PFTTrDA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2058-94-8	PFUnA	ND	1.7	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C2-4:2FTS			298 %	40-200		11/04/25	B5J3030	EPA 1633	DT	IDA01
Isotope Dilution Analog: 13C2-6:2FTS			218 %	40-200		11/04/25	B5J3030	EPA 1633	DT	IDA01
Isotope Dilution Analog: 13C2-8:2FTS			200 %	40-300		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C2-PFDA			155 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C2-PFDoA			67.3 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA			61.6 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C2-PFHxA			139 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C2-PFTeDA			51.3 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C3-HFPODA			80.0 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C3-PFBA			136 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C3-PFBS			88.2 %	40-135		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C3-PFHxS			86.9 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C3-PFPrA			36.8 %	20-150		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C4-PFBA			86.0 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C4-PFHpA			90.5 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C4-PFOA			146 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C4-PFOS			143 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C5-PFHxA			87.2 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 13C5-PFNA			148 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C5-PFPeA			83.5 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C6-PFDA			81.6 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C7-PFUnA			82.5 %	30-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C8-PFOA			86.3 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C8-PFOS			78.6 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C8-PFOSA			95.0 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: 13C9-PFNA			88.2 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Performance Standard: 18O2-PFHxS			136 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d3-NMeFOSA			70.2 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d3-N-MeFOSAA			69.1 %	40-170		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d5-NEtFOSA			55.6 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d5-N-EtFOSAA			95.0 %	25-135		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d7-NMeFOSE			90.7 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
Isotope Dilution Analog: d9-NEtFOSE			79.1 %	10-130		11/04/25	B5J3030	EPA 1633	DT	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Client ID: TB
Lab ID: 2510354-02

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Analyst	Qualifier
Organics-Semivolatiles										
763051-92-9	11Cl-PF3OUdS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
356-02-5	3:3FTCA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
757124-72-4	4:2FTS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
914637-49-3	5:3FTCA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
27619-97-2	6:2FTS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
812-70-4	7:3FTCA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
39108-34-4	8:2FTS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
756426-58-1	9Cl-PF3ONS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
919005-14-4	ADONA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
13252-13-6	HFPO-DA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
4151-50-2	NEtFOSA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2991-50-6	NEtFOSAA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
1691-99-2	NEtFOSE	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
151772-58-6	NFDHA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
31506-32-8	NMeFOSA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2355-31-9	NMeFOSAA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
24448-09-7	NMeFOSE	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-22-4	PFBA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-73-5	PFBS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
30334-69-1	PFBSA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
335-76-2	PFDA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
307-55-1	PFDaA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
79780-39-5	PFDoS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
335-77-3	PFDS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
646-83-3	PFECHS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
113507-82-7	PFEESA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-85-9	PFHpA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-92-8	PFHpS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
307-24-4	PFHxA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
355-46-4	PFHxS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
41997-13-1	PFHxSA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
863090-89-5	PFMBA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
377-73-1	PFMPA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
375-95-1	PFNA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
68259-12-1	PFNS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
335-67-1	PFOA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
1763-23-1	PFOS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
754-91-6	PFOSA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2706-90-3	PFPeA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2706-91-4	PFPeS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
422-64-0	PFPrA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
423-41-6	PFPrS	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Client ID: TB
Lab ID: 2510354-02

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Analyst	Qualifier
Organics-Semivolatiles										
376-06-7	PFTeDA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
72629-94-8	PFTTrDA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
2058-94-8	PFUnA	ND	1.9	ng/L	1	11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-4:2FTS</i>			94.6 %	40-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-6:2FTS</i>			84.2 %	40-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-8:2FTS</i>			89.2 %	40-300		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C2-PFDA</i>			135 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-PFDoA</i>			76.6 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA</i>			79.7 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C2-PFHxA</i>			129 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C2-PFTeDA</i>			85.2 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C3-HFPODA</i>			83.4 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C3-PFBA</i>			127 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C3-PFBS</i>			84.3 %	40-135		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C3-PFHxS</i>			79.5 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C3-PFPrA</i>			111 %	20-150		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C4-PFBA</i>			87.5 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C4-PFHpA</i>			85.3 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C4-PFOA</i>			129 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C4-PFOS</i>			130 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C5-PFHxA</i>			88.3 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 13C5-PFNA</i>			127 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C5-PFPeA</i>			87.8 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C6-PFDA</i>			74.9 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C7-PFUnA</i>			76.0 %	30-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C8-PFOA</i>			81.9 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C8-PFOS</i>			75.9 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C8-PFOSA</i>			101 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: 13C9-PFNA</i>			78.5 %	40-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Performance Standard: 18O2-PFHxS</i>			124 %	50-200		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d3-NMeFOSA</i>			81.0 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d3-N-MeFOSAA</i>			56.0 %	40-170		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d5-NEtFOSA</i>			68.0 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d5-N-EtFOSAA</i>			75.3 %	25-135		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d7-NMeFOSE</i>			102 %	10-130		11/04/25	B5J3030	EPA 1633	DT	
<i>Isotope Dilution Analog: d9-NEtFOSE</i>			90.9 %	10-130		11/04/25	B5J3030	EPA 1633	DT	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	----------------	------------------	------	----------------	-----	--------------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Blank (B5J3030-BLK1)

11Cl-PF3OUdS	ND	ng/L							11/04/2025	
3:3FTCA	ND	ng/L							11/04/2025	
4:2FTS	ND	ng/L							11/04/2025	
5:3FTCA	ND	ng/L							11/04/2025	
6:2FTS	ND	ng/L							11/04/2025	
7:3FTCA	ND	ng/L							11/04/2025	
8:2FTS	ND	ng/L							11/04/2025	
9Cl-PF3ONS	ND	ng/L							11/04/2025	
ADONA	ND	ng/L							11/04/2025	
HFPO-DA	ND	ng/L							11/04/2025	
NEtFOSA	ND	ng/L							11/04/2025	
NEtFOSAA	ND	ng/L							11/04/2025	
NEtFOSE	ND	ng/L							11/04/2025	
NFDHA	ND	ng/L							11/04/2025	
NMeFOSA	ND	ng/L							11/04/2025	
NMeFOSAA	ND	ng/L							11/04/2025	
NMeFOSE	ND	ng/L							11/04/2025	
PFBA	ND	ng/L							11/04/2025	
PFBS	ND	ng/L							11/04/2025	
PFBSA	ND	ng/L							11/04/2025	
PFDA	ND	ng/L							11/04/2025	
PFDaA	ND	ng/L							11/04/2025	
PFDoS	ND	ng/L							11/04/2025	
PFDS	ND	ng/L							11/04/2025	
PFECHS	ND	ng/L							11/04/2025	
PFEESA	ND	ng/L							11/04/2025	
PFHpA	ND	ng/L							11/04/2025	
PFHpS	ND	ng/L							11/04/2025	
PFHxA	ND	ng/L							11/04/2025	
PFHxS	ND	ng/L							11/04/2025	
PFHxSA	ND	ng/L							11/04/2025	
PFMBA	ND	ng/L							11/04/2025	
PFMPA	ND	ng/L							11/04/2025	
PFNA	ND	ng/L							11/04/2025	
PFNS	ND	ng/L							11/04/2025	
PFOA	ND	ng/L							11/04/2025	
PFOS	ND	ng/L							11/04/2025	
PFOSA	ND	ng/L							11/04/2025	
PFPeA	ND	ng/L							11/04/2025	
PFPeS	ND	ng/L							11/04/2025	
PFPPrA	ND	ng/L							11/04/2025	
PFPPrS	ND	ng/L							11/04/2025	
PFTeDA	ND	ng/L							11/04/2025	
PFTTrDA	ND	ng/L							11/04/2025	
PFUnA	ND	ng/L							11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	32.7	ng/L	37.52		87.2	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-6:2FTS	31.7	ng/L	38.04		83.4	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	31.7	ng/L	38.40		82.6	40-300			11/04/2025	
Isotope Performance Standard: 13C2-PFDA	1740	ng/L	1250		139	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDaA	7.87	ng/L	10.00		78.7	10-130			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	----------------	------------------	------	----------------	-----	--------------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Blank (B5J3030-BLK1)

Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA	7.32	ng/L	10.00		73.2	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3450	ng/L	2500		138	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	6.34	ng/L	10.00		63.4	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	65.4	ng/L	80.00		81.8	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6860	ng/L	5000		137	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	15.1	ng/L	18.64		80.8	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	14.3	ng/L	18.96		75.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPrA	85.8	ng/L	80.00		107	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	67.2	ng/L	80.00		84.1	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHpA	16.7	ng/L	20.00		83.5	40-130			11/04/2025	
Isotope Performance Standard: 13C4-PFOA	3430	ng/L	2500		137	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3190	ng/L	2395		133	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFHxA	16.6	ng/L	20.00		82.8	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1730	ng/L	1250		138	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	34.4	ng/L	40.00		86.1	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	8.50	ng/L	10.00		85.0	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	7.89	ng/L	10.00		78.9	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	16.1	ng/L	20.00		80.6	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	14.6	ng/L	19.16		76.0	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	15.9	ng/L	20.00		79.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	7.92	ng/L	10.00		79.2	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3190	ng/L	2370		135	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	13.0	ng/L	20.00		65.1	10-130			11/04/2025	
Isotope Dilution Analog: d3-N-MeFOSAA	25.7	ng/L	40.00		64.2	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	12.5	ng/L	20.00		62.7	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	33.2	ng/L	40.00		83.0	25-135			11/04/2025	
Isotope Dilution Analog: d7-NMeFOSE	156	ng/L	200.0		78.2	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	157	ng/L	200.0		78.6	10-130			11/04/2025	

LCS (B5J3030-BS1)

11Cl-PF3OUdS	30.2	ng/L	37.80		79.8	55-160			11/04/2025	
3:3FTCA	34.9	ng/L	40.00		87.2	65-130			11/04/2025	
4:2FTS	34.8	ng/L	37.40		93.1	70-145			11/04/2025	
5:3FTCA	36.1	ng/L	40.00		90.3	70-135			11/04/2025	
6:2FTS	36.2	ng/L	38.00		95.1	65-155			11/04/2025	
7:3FTCA	31.4	ng/L	40.00		78.5	50-145			11/04/2025	
8:2FTS	40.6	ng/L	38.40		106	60-150			11/04/2025	
9Cl-PF3ONS	29.6	ng/L	37.40		79.2	70-155			11/04/2025	
ADONA	33.0	ng/L	37.80		87.2	65-145			11/04/2025	
HFPO-DA	33.1	ng/L	40.00		82.9	70-140			11/04/2025	
NEtFOSA	35.2	ng/L	40.00		88.1	65-145			11/04/2025	
NEtFOSAA	35.5	ng/L	40.00		88.7	70-145			11/04/2025	
NEtFOSE	36.2	ng/L	40.00		90.5	70-135			11/04/2025	
NFDHA	42.0	ng/L	40.00		105	50-150			11/04/2025	
NMeFOSA	33.3	ng/L	40.00		83.2	60-150			11/04/2025	
NMeFOSAA	35.9	ng/L	40.00		89.8	50-140			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

LCS (B5J3030-BS1)

NMeFOSE	35.2	ng/L	40.00		87.9	70-145			11/04/2025	
PFBA	35.4	ng/L	40.00		88.4	70-140			11/04/2025	
PFBS	30.5	ng/L	35.40		86.2	60-145			11/04/2025	
PFBSA	36.1	ng/L	40.00		90.2	60-140			11/04/2025	
PFDA	33.7	ng/L	40.00		84.4	70-140			11/04/2025	
PFDoA	36.2	ng/L	40.00		90.4	70-140			11/04/2025	
PFDoS	30.0	ng/L	38.80		77.3	50-145			11/04/2025	
PFDS	35.1	ng/L	38.60		90.8	60-145			11/04/2025	
PFECHS	34.1	ng/L	36.96		92.3	60-140			11/04/2025	
PFEESA	30.5	ng/L	35.60		85.7	70-140			11/04/2025	
PFHpA	36.2	ng/L	40.00		90.6	70-150			11/04/2025	
PFHpS	36.0	ng/L	38.20		94.3	70-150			11/04/2025	
PFHxA	36.4	ng/L	40.00		91.0	70-145			11/04/2025	
PFHxS	31.6	ng/L	36.50		86.5	65-145			11/04/2025	
PFHxSA	36.8	ng/L	40.00		92.1	60-140			11/04/2025	
PFMBA	32.7	ng/L	40.00		81.8	60-150			11/04/2025	
PFMPA	36.1	ng/L	40.00		90.2	55-140			11/04/2025	
PFNA	36.0	ng/L	40.00		89.9	70-150			11/04/2025	
PFNS	33.7	ng/L	38.40		87.9	65-145			11/04/2025	
PFOA	35.8	ng/L	40.00		89.4	70-150			11/04/2025	
PFOS	31.6	ng/L	37.04		85.4	55-150			11/04/2025	
PFOSA	34.2	ng/L	40.00		85.4	70-145			11/04/2025	
PFPeA	35.8	ng/L	40.00		89.6	65-135			11/04/2025	
PFPeS	34.2	ng/L	37.60		91.0	65-140			11/04/2025	
PFPrA	32.9	ng/L	40.00		82.3	60-140			11/04/2025	
PFPrS	31.9	ng/L	36.80		86.6	60-140			11/04/2025	
PFTeDA	36.0	ng/L	40.00		90.1	60-140			11/04/2025	
PFTrDA	37.0	ng/L	40.00		92.6	65-140			11/04/2025	
PFUnA	35.0	ng/L	40.00		87.6	70-145			11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	33.1	ng/L	37.52		88.2	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-6:2FTS	31.9	ng/L	38.04		83.9	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	30.4	ng/L	38.40		79.2	40-300			11/04/2025	
Isotope Performance Standard: 13C2-PFDA	1670	ng/L	1250		133	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA	8.19	ng/L	10.00		81.9	10-130			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA	7.93	ng/L	10.00		79.3	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3320	ng/L	2500		133	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	7.46	ng/L	10.00		74.6	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	70.0	ng/L	80.00		87.5	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6600	ng/L	5000		132	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	16.1	ng/L	18.64		86.2	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	16.3	ng/L	18.96		85.8	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPrA	91.9	ng/L	80.00		115	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	72.5	ng/L	80.00		90.6	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHpA	17.4	ng/L	20.00		87.2	40-130			11/04/2025	
Isotope Performance Standard: 13C4-PFOA	3290	ng/L	2500		131	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3060	ng/L	2395		128	50-200			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

LCS (B5J3030-BS1)

Isotope Dilution Analog: 13C5-PFHxA	17.8	ng/L	20.00		88.9	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1600	ng/L	1250		128	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	36.7	ng/L	40.00		91.8	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	8.55	ng/L	10.00		85.5	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	8.31	ng/L	10.00		83.1	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	17.7	ng/L	20.00		88.5	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	15.6	ng/L	19.16		81.7	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	17.3	ng/L	20.00		86.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	9.07	ng/L	10.00		90.7	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3080	ng/L	2370		130	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	14.7	ng/L	20.00		73.7	10-130			11/04/2025	
Isotope Dilution Analog: d3-N-MeFOSAA	26.3	ng/L	40.00		65.7	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	13.1	ng/L	20.00		65.7	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	34.5	ng/L	40.00		86.3	25-135			11/04/2025	
Isotope Dilution Analog: d7-NMeFOSE	171	ng/L	200.0		85.5	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	167	ng/L	200.0		83.4	10-130			11/04/2025	

LCS Dup (B5J3030-BSD1)

11CI-PF3OUdS	31.2	ng/L	37.80		82.6	55-160	3.45	30	11/04/2025	
3:3FTCA	35.7	ng/L	40.00		89.3	65-130	2.33	30	11/04/2025	
4:2FTS	36.4	ng/L	37.40		97.2	70-145	4.33	30	11/04/2025	
5:3FTCA	37.4	ng/L	40.00		93.4	70-135	3.36	30	11/04/2025	
6:2FTS	37.0	ng/L	38.00		97.4	65-155	2.34	30	11/04/2025	
7:3FTCA	33.8	ng/L	40.00		84.4	50-145	7.24	30	11/04/2025	
8:2FTS	40.0	ng/L	38.40		104	60-150	1.26	30	11/04/2025	
9CI-PF3ONS	30.4	ng/L	37.40		81.2	70-155	2.55	30	11/04/2025	
ADONA	35.1	ng/L	37.80		93.0	65-145	6.34	30	11/04/2025	
HFPO-DA	32.5	ng/L	40.00		81.2	70-140	1.98	30	11/04/2025	
NEtFOSA	34.9	ng/L	40.00		87.1	65-145	1.11	30	11/04/2025	
NEtFOSAA	35.0	ng/L	40.00		87.6	70-145	1.29	30	11/04/2025	
NEtFOSE	36.7	ng/L	40.00		91.8	70-135	1.47	30	11/04/2025	
NFDHA	41.7	ng/L	40.00		104	50-150	0.706	30	11/04/2025	
NMeFOSA	35.4	ng/L	40.00		88.6	60-150	6.23	30	11/04/2025	
NMeFOSAA	37.2	ng/L	40.00		92.9	50-140	3.42	30	11/04/2025	
NMeFOSE	36.7	ng/L	40.00		91.7	70-145	4.17	30	11/04/2025	
PFBA	36.5	ng/L	40.00		91.2	70-140	3.08	30	11/04/2025	
PFBS	31.1	ng/L	35.40		87.7	60-145	1.82	30	11/04/2025	
PFBSA	37.7	ng/L	40.00		94.3	60-140	4.41	30	11/04/2025	
PFDA	36.1	ng/L	40.00		90.3	70-140	6.77	30	11/04/2025	
PFDoA	36.0	ng/L	40.00		89.9	70-140	0.538	30	11/04/2025	
PFDoS	29.1	ng/L	38.80		74.9	50-145	3.07	30	11/04/2025	
PFDS	35.8	ng/L	38.60		92.7	60-145	2.02	30	11/04/2025	
PFECHS	34.5	ng/L	36.96		93.2	60-140	1.01	30	11/04/2025	
PFEESA	31.7	ng/L	35.60		89.1	70-140	3.86	30	11/04/2025	
PFHpA	36.0	ng/L	40.00		90.0	70-150	0.604	30	11/04/2025	
PFHpS	36.0	ng/L	38.20		94.2	70-150	0.124	30	11/04/2025	
PFHxA	36.7	ng/L	40.00		91.8	70-145	0.816	30	11/04/2025	
PFHxS	32.9	ng/L	36.50		90.2	65-145	4.14	30	11/04/2025	
PFHxSA	38.8	ng/L	40.00		97.1	60-140	5.30	30	11/04/2025	
PFMBA	33.1	ng/L	40.00		82.7	60-150	1.10	30	11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	----------------	------------------	------	----------------	-----	--------------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

LCS Dup (B5J3030-BSD1)

PFMPA	37.0	ng/L	40.00		92.5	55-140	2.52	30	11/04/2025	
PFNA	37.7	ng/L	40.00		94.3	70-150	4.82	30	11/04/2025	
PFNS	34.9	ng/L	38.40		91.0	65-145	3.49	30	11/04/2025	
PFOA	36.0	ng/L	40.00		89.9	70-150	0.608	30	11/04/2025	
PFOS	32.9	ng/L	37.04		88.9	55-150	4.02	30	11/04/2025	
PFOSA	35.3	ng/L	40.00		88.3	70-145	3.28	30	11/04/2025	
PFPeA	36.9	ng/L	40.00		92.2	65-135	2.94	30	11/04/2025	
PFPeS	36.4	ng/L	37.60		96.7	65-140	6.13	30	11/04/2025	
PFPPrA	33.5	ng/L	40.00		83.6	60-140	1.61	30	11/04/2025	
PFPPrS	32.5	ng/L	36.80		88.4	60-140	2.02	30	11/04/2025	
PFTeDA	37.5	ng/L	40.00		93.7	60-140	3.94	30	11/04/2025	
PFTTrDA	36.6	ng/L	40.00		91.4	65-140	1.28	30	11/04/2025	
PFUnA	34.6	ng/L	40.00		86.5	70-145	1.18	30	11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	32.4	ng/L	37.52		86.3	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-6:2FTS	32.4	ng/L	38.04		85.3	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	28.3	ng/L	38.40		73.8	40-300			11/04/2025	
Isotope Performance Standard: 13C2-PFDA	1720	ng/L	1250		138	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDaA	7.87	ng/L	10.00		78.7	10-130			11/04/2025	
Isotope Dilution Analog: 13C2-PFDaA+13C2-PFTeDA	7.44	ng/L	10.00		74.4	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3380	ng/L	2500		135	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	6.65	ng/L	10.00		66.5	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	65.3	ng/L	80.00		81.6	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6700	ng/L	5000		134	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	15.8	ng/L	18.64		84.5	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	15.0	ng/L	18.96		79.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPPrA	89.9	ng/L	80.00		112	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	70.7	ng/L	80.00		88.4	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHpA	17.1	ng/L	20.00		85.6	40-130			11/04/2025	
Isotope Performance Standard: 13C4-PFOA	3360	ng/L	2500		134	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3050	ng/L	2395		128	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFHxA	17.1	ng/L	20.00		85.6	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1590	ng/L	1250		127	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	35.7	ng/L	40.00		89.1	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	7.69	ng/L	10.00		76.9	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	7.86	ng/L	10.00		78.6	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	16.7	ng/L	20.00		83.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	14.8	ng/L	19.16		77.2	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	16.3	ng/L	20.00		81.5	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	8.54	ng/L	10.00		85.4	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3140	ng/L	2370		133	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	13.7	ng/L	20.00		68.4	10-130			11/04/2025	
Isotope Dilution Analog: d3-N-MeFOSAA	24.3	ng/L	40.00		60.9	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	12.8	ng/L	20.00		63.8	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	33.8	ng/L	40.00		84.5	25-135			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

LCS Dup (B5J3030-BSD1)

Isotope Dilution Analog: d7-NMeFOSE	162	ng/L	200.0		80.8	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	162	ng/L	200.0		81.2	10-130			11/04/2025	

MRL Check (B5J3030-MRL1)

11Cl-PF3OUdS	3.20	ng/L	3.780		84.6	55-160			11/04/2025	
3:3FTCA	5.02	ng/L	4.000		125	65-130			11/04/2025	
4:2FTS	4.02	ng/L	3.740		107	70-145			11/04/2025	
5:3FTCA	5.18	ng/L	4.000		130	70-135			11/04/2025	
6:2FTS	4.16	ng/L	3.800		109	65-155			11/04/2025	
7:3FTCA	3.99	ng/L	4.000		99.6	50-145			11/04/2025	
8:2FTS	4.68	ng/L	3.840		122	60-150			11/04/2025	
9Cl-PF3ONS	3.40	ng/L	3.740		90.9	70-155			11/04/2025	
ADONA	3.76	ng/L	3.780		99.4	65-145			11/04/2025	
HFPO-DA	4.07	ng/L	4.000		102	70-140			11/04/2025	
NEtFOSA	3.90	ng/L	4.000		97.5	65-145			11/04/2025	
NEtFOSAA	4.21	ng/L	4.000		105	70-145			11/04/2025	
NEtFOSE	4.18	ng/L	4.000		104	70-135			11/04/2025	
NFDHA	4.84	ng/L	4.000		121	50-150			11/04/2025	
NMeFOSA	3.65	ng/L	4.000		91.1	60-150			11/04/2025	
NMeFOSAA	3.81	ng/L	4.000		95.2	50-140			11/04/2025	
NMeFOSE	4.40	ng/L	4.000		110	70-145			11/04/2025	
PFBA	4.16	ng/L	4.000		104	70-140			11/04/2025	
PFBS	3.63	ng/L	3.540		103	60-145			11/04/2025	
PFBSA	4.20	ng/L	4.000		105	60-140			11/04/2025	
PFDA	3.81	ng/L	4.000		95.1	70-140			11/04/2025	
PFDoA	4.14	ng/L	4.000		103	70-140			11/04/2025	
PFDoS	3.30	ng/L	3.880		85.2	50-145			11/04/2025	
PFDS	3.54	ng/L	3.860		91.6	60-145			11/04/2025	
PFECHS	4.05	ng/L	3.696		110	60-140			11/04/2025	
PFEESA	3.49	ng/L	3.560		98.1	70-140			11/04/2025	
PFHpA	4.13	ng/L	4.000		103	70-150			11/04/2025	
PFHpS	4.44	ng/L	3.820		116	70-150			11/04/2025	
PFHxA	4.20	ng/L	4.000		105	70-145			11/04/2025	
PFHxS	3.65	ng/L	3.650		99.9	65-145			11/04/2025	
PFHxSA	4.45	ng/L	4.000		111	60-140			11/04/2025	
PFMBA	3.97	ng/L	4.000		99.3	60-150			11/04/2025	
PFMPA	4.44	ng/L	4.000		111	55-140			11/04/2025	
PFNA	4.22	ng/L	4.000		105	70-150			11/04/2025	
PFNS	4.00	ng/L	3.840		104	65-145			11/04/2025	
PFOA	4.01	ng/L	4.000		100	70-150			11/04/2025	
PFOS	4.23	ng/L	3.704		114	55-150			11/04/2025	
PFOSA	3.95	ng/L	4.000		98.8	70-145			11/04/2025	
PFPeA	4.13	ng/L	4.000		103	65-135			11/04/2025	
PFPeS	3.90	ng/L	3.760		104	65-140			11/04/2025	
PFPPrA	3.73	ng/L	4.000		93.4	60-140			11/04/2025	
PFPPrS	3.78	ng/L	3.680		103	60-140			11/04/2025	
PFTeDA	4.09	ng/L	4.000		102	60-140			11/04/2025	
PFTTrDA	4.00	ng/L	4.000		100	65-140			11/04/2025	
PFUnA	4.09	ng/L	4.000		102	70-145			11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	34.6	ng/L	37.52		92.3	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-6:2FTS	32.7	ng/L	38.04		85.9	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	31.2	ng/L	38.40		81.2	40-300			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

MRL Check (B5J3030-MRL1)

Isotope Performance Standard: 13C2-PFDA	1720	ng/L	1250		137	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA	7.72	ng/L	10.00		77.2	10-130			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA	7.16	ng/L	10.00		71.6	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3240	ng/L	2500		130	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	6.14	ng/L	10.00		61.4	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	69.7	ng/L	80.00		87.1	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6500	ng/L	5000		130	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	15.7	ng/L	18.64		84.3	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	15.5	ng/L	18.96		81.5	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPrA	91.6	ng/L	80.00		115	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	72.7	ng/L	80.00		90.9	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHpA	17.8	ng/L	20.00		89.2	40-130			11/04/2025	
Isotope Performance Standard: 13C4-PFOA	3320	ng/L	2500		133	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3140	ng/L	2395		131	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFHxA	17.9	ng/L	20.00		89.5	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1590	ng/L	1250		127	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	37.0	ng/L	40.00		92.6	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	8.08	ng/L	10.00		80.8	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	7.64	ng/L	10.00		76.4	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	16.8	ng/L	20.00		83.8	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	13.9	ng/L	19.16		72.6	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	16.4	ng/L	20.00		82.0	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	8.74	ng/L	10.00		87.4	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3070	ng/L	2370		130	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	13.4	ng/L	20.00		66.9	10-130			11/04/2025	
Isotope Dilution Analog: d3-N-MeFOSAA	25.0	ng/L	40.00		62.4	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	12.2	ng/L	20.00		61.2	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	31.7	ng/L	40.00		79.4	25-135			11/04/2025	
Isotope Dilution Analog: d7-NMeFOSE	153	ng/L	200.0		76.3	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	153	ng/L	200.0		76.4	10-130			11/04/2025	

Matrix Spike (B5J3030-MS1)

Source: 2510354-01

11CI-PF3OUdS	28.6	ng/L	32.86	ND	87.0	55-160			11/04/2025	
3:3FTCA	44.3	ng/L	34.77	0.459	126	65-130			11/04/2025	
4:2FTS	32.4	ng/L	32.51	ND	99.5	70-145			11/04/2025	IDA01
5:3FTCA	37.5	ng/L	34.77	1.67	103	70-135			11/04/2025	
6:2FTS	33.8	ng/L	33.04	0.291	102	65-155			11/04/2025	
7:3FTCA	34.9	ng/L	34.77	ND	100	50-145			11/04/2025	
8:2FTS	36.4	ng/L	33.38	ND	109	60-150			11/04/2025	
9CI-PF3ONS	30.5	ng/L	32.51	ND	93.8	70-155			11/04/2025	
ADONA	33.5	ng/L	32.86	0.0950	102	65-145			11/04/2025	
HFPO-DA	32.9	ng/L	34.77	ND	94.6	70-140			11/04/2025	
NEtFOSA	32.7	ng/L	34.77	ND	94.0	65-145			11/04/2025	
NEtFOSAA	30.1	ng/L	34.77	ND	86.7	70-145			11/04/2025	
NEtFOSE	32.1	ng/L	34.77	ND	92.3	70-135			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Matrix Spike (B5J3030-MS1)

Source: 2510354-01

NFDHA	30.9	ng/L	34.77	ND	89.0	50-150			11/04/2025	
NMeFOSA	33.0	ng/L	34.77	ND	95.0	60-150			11/04/2025	
NMeFOSAA	32.4	ng/L	34.77	0.574	91.4	50-140			11/04/2025	
NMeFOSE	31.9	ng/L	34.77	ND	91.8	70-145			11/04/2025	
PFBA	34.6	ng/L	34.77	1.33	95.7	70-140			11/04/2025	
PFBS	33.7	ng/L	30.78	4.41	95.2	60-145			11/04/2025	
PFBSA	33.3	ng/L	34.77	1.33	92.0	60-140			11/04/2025	
PFDA	31.8	ng/L	34.77	0.256	90.6	70-140			11/04/2025	
PFDoA	30.3	ng/L	34.77	ND	87.0	70-140			11/04/2025	
PFDoS	11.8	ng/L	33.73	ND	34.9	50-145			11/04/2025	A03
PFDS	24.0	ng/L	33.56	ND	71.5	60-145			11/04/2025	
PFECHS	27.9	ng/L	32.13	0.121	86.3	60-140			11/04/2025	
PFEESA	28.8	ng/L	30.95	ND	92.9	70-140			11/04/2025	
PFHpA	32.3	ng/L	34.77	0.339	92.0	70-150			11/04/2025	
PFHpS	31.2	ng/L	33.21	ND	94.0	70-150			11/04/2025	
PFHxA	39.0	ng/L	34.77	6.27	94.0	70-145			11/04/2025	
PFHxS	30.6	ng/L	31.73	0.303	95.6	65-145			11/04/2025	
PFHxSA	32.5	ng/L	34.77	ND	93.6	60-140			11/04/2025	
PFMBA	31.9	ng/L	34.77	ND	91.7	60-150			11/04/2025	
PFMPA	34.6	ng/L	34.77	0.539	97.9	55-140			11/04/2025	
PFNA	32.6	ng/L	34.77	0.192	93.2	70-150			11/04/2025	
PFNS	27.4	ng/L	33.38	ND	82.2	65-145			11/04/2025	
PFOA	33.4	ng/L	34.77	1.36	92.2	70-150			11/04/2025	
PFOS	29.1	ng/L	32.20	0.908	87.6	55-150			11/04/2025	
PFOSA	31.3	ng/L	34.77	0.145	89.7	70-145			11/04/2025	
PFPeA	34.7	ng/L	34.77	1.57	95.4	65-135			11/04/2025	
PFPeS	31.1	ng/L	32.69	ND	95.1	65-140			11/04/2025	
PFPrA	45.7	ng/L	34.77	16.0	85.5	60-140			11/04/2025	
PFPrS	29.2	ng/L	31.99	0.188	90.8	60-140			11/04/2025	
PFTeDA	35.2	ng/L	34.77	ND	101	60-140			11/04/2025	
PFTTrDA	29.0	ng/L	34.77	ND	83.5	65-140			11/04/2025	
PFUnA	29.9	ng/L	34.77	ND	86.1	70-145			11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	97.7	ng/L	32.62		300	40-200			11/04/2025	IDA01
Isotope Dilution Analog: 13C2-6:2FTS	63.9	ng/L	33.07		193	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	55.7	ng/L	33.38		167	40-300			11/04/2025	
Isotope Performance Standard: 13C2-PFDA	2020	ng/L	1250		161	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA	6.39	ng/L	8.694		73.5	10-130			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA	5.80	ng/L	8.694		66.8	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3560	ng/L	2500		143	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	4.75	ng/L	8.694		54.6	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	53.3	ng/L	69.55		76.6	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6970	ng/L	5000		139	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	14.3	ng/L	16.20		88.3	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	14.3	ng/L	16.48		86.5	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPrA	26.1	ng/L	69.55		37.6	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	60.3	ng/L	69.55		86.7	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHxA	15.5	ng/L	17.39		89.1	40-130			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Matrix Spike (B5J3030-MS1)

Source: 2510354-01

Isotope Performance Standard: 13C4-PFOA	3780	ng/L	2500		151	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3510	ng/L	2395		147	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFHxA	15.1	ng/L	17.39		86.6	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1890	ng/L	1250		151	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	28.6	ng/L	34.77		82.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	7.00	ng/L	8.694		80.6	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	7.24	ng/L	8.694		83.3	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	15.2	ng/L	17.39		87.3	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	13.7	ng/L	16.66		82.2	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	18.6	ng/L	17.39		107	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	7.78	ng/L	8.694		89.4	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3310	ng/L	2370		139	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	13.7	ng/L	17.39		78.9	10-130			11/04/2025	
Isotope Dilution Analog: d3-N-MeFOSAA	25.0	ng/L	34.77		71.9	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	11.0	ng/L	17.39		63.3	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	35.5	ng/L	34.77		102	25-135			11/04/2025	
Isotope Dilution Analog: d7-NMeFOSE	173	ng/L	173.9		99.4	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	144	ng/L	173.9		82.8	10-130			11/04/2025	

Matrix Spike Dup (B5J3030-MSD1)

Source: 2510354-01

11Cl-PF3OUdS	27.2	ng/L	33.01	ND	82.5	55-160	4.83	30	11/04/2025	
3:3FTCA	43.6	ng/L	34.93	0.459	124	65-130	1.51	30	11/04/2025	
4:2FTS	31.8	ng/L	32.66	ND	97.2	70-145	1.89	30	11/04/2025	IDA01
5:3FTCA	35.7	ng/L	34.93	1.67	97.5	70-135	4.83	30	11/04/2025	
6:2FTS	32.3	ng/L	33.18	0.291	96.5	65-155	4.62	30	11/04/2025	
7:3FTCA	33.4	ng/L	34.93	ND	95.5	50-145	4.46	30	11/04/2025	
8:2FTS	34.7	ng/L	33.53	ND	104	60-150	4.76	30	11/04/2025	
9Cl-PF3ONS	27.6	ng/L	32.66	ND	84.4	70-155	10.1	30	11/04/2025	
ADONA	30.6	ng/L	33.01	0.0950	92.4	65-145	8.97	30	11/04/2025	
HFPO-DA	32.2	ng/L	34.93	ND	92.3	70-140	2.05	30	11/04/2025	
NEtFOSA	33.1	ng/L	34.93	ND	94.8	65-145	1.36	30	11/04/2025	
NEtFOSAA	31.5	ng/L	34.93	ND	90.3	70-145	4.54	30	11/04/2025	
NEtFOSE	30.3	ng/L	34.93	ND	86.9	70-135	5.64	30	11/04/2025	
NFDHA	26.4	ng/L	34.93	ND	75.6	50-150	15.9	30	11/04/2025	
NMeFOSA	32.4	ng/L	34.93	ND	92.7	60-150	2.01	30	11/04/2025	
NMeFOSAA	31.7	ng/L	34.93	0.574	89.3	50-140	1.92	30	11/04/2025	
NMeFOSE	30.6	ng/L	34.93	ND	87.6	70-145	4.26	30	11/04/2025	
PFBA	33.7	ng/L	34.93	1.33	92.6	70-140	2.76	30	11/04/2025	
PFBS	33.0	ng/L	30.91	4.41	92.3	60-145	2.32	30	11/04/2025	
PFBSA	32.6	ng/L	34.93	1.33	89.5	60-140	2.20	30	11/04/2025	
PFDA	30.3	ng/L	34.93	0.256	86.1	70-140	4.66	30	11/04/2025	
PFDoA	31.4	ng/L	34.93	ND	90.0	70-140	3.79	30	11/04/2025	
PFDoS	13.2	ng/L	33.88	ND	39.0	50-145	11.4	30	11/04/2025	A03
PFDS	26.5	ng/L	33.70	ND	78.5	60-145	9.79	30	11/04/2025	
PFECHS	28.1	ng/L	32.27	0.121	86.7	60-140	0.833	30	11/04/2025	
PFEESA	28.0	ng/L	31.09	ND	90.0	70-140	2.71	30	11/04/2025	
PFHpA	31.5	ng/L	34.93	0.339	89.3	70-150	2.43	30	11/04/2025	
PFHpS	30.0	ng/L	33.36	ND	89.8	70-150	4.13	30	11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
ENVIRONMENTAL LABORATORY

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Matrix Spike Dup (B5J3030-MSD1)

Source: 2510354-01

PFHxA	38.2	ng/L	34.93	6.27	91.5	70-145	1.90	30	11/04/2025	
PFHxS	29.5	ng/L	31.87	0.303	91.8	65-145	3.66	30	11/04/2025	
PFHxSA	32.6	ng/L	34.93	ND	93.4	60-140	0.226	30	11/04/2025	
PFMBA	31.0	ng/L	34.93	ND	88.8	60-150	2.73	30	11/04/2025	
PFMPA	33.9	ng/L	34.93	0.539	95.4	55-140	2.08	30	11/04/2025	
PFNA	34.4	ng/L	34.93	0.192	98.0	70-150	5.43	30	11/04/2025	
PFNS	26.3	ng/L	33.53	ND	78.4	65-145	4.22	30	11/04/2025	
PFOA	32.8	ng/L	34.93	1.36	90.0	70-150	1.93	30	11/04/2025	
PFOS	28.3	ng/L	32.34	0.908	84.7	55-150	2.78	30	11/04/2025	
PFOSA	31.2	ng/L	34.93	0.145	89.0	70-145	0.308	30	11/04/2025	
PFPeA	33.8	ng/L	34.93	1.57	92.2	65-135	2.78	30	11/04/2025	
PFPeS	29.9	ng/L	32.83	ND	91.2	65-140	3.71	30	11/04/2025	
PFPrA	46.0	ng/L	34.93	16.0	85.9	60-140	0.592	30	11/04/2025	
PFPrS	28.1	ng/L	32.13	0.188	86.7	60-140	4.05	30	11/04/2025	
PFTeDA	32.4	ng/L	34.93	ND	92.7	60-140	8.32	30	11/04/2025	
PFTrDA	30.1	ng/L	34.93	ND	86.1	65-140	3.45	30	11/04/2025	
PFUnA	30.8	ng/L	34.93	ND	88.1	70-145	2.78	30	11/04/2025	
Isotope Dilution Analog: 13C2-4:2FTS	96.1	ng/L	32.76		293	40-200			11/04/2025	IDA01
Isotope Dilution Analog: 13C2-6:2FTS	64.5	ng/L	33.22		194	40-200			11/04/2025	
Isotope Dilution Analog: 13C2-8:2FTS	58.0	ng/L	33.53		173	40-300			11/04/2025	
Isotope Performance Standard: 13C2-PFDA	1910	ng/L	1250		152	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA	6.57	ng/L	8.732		75.2	10-130			11/04/2025	
Isotope Dilution Analog: 13C2-PFDoA+13C2-PFTeDA	6.19	ng/L	8.732		70.9	10-130			11/04/2025	
Isotope Performance Standard: 13C2-PFHxA	3420	ng/L	2500		137	50-200			11/04/2025	
Isotope Dilution Analog: 13C2-PFTeDA	5.51	ng/L	8.732		63.1	10-130			11/04/2025	
Isotope Dilution Analog: 13C3-HFPODA	59.4	ng/L	69.85		85.0	40-130			11/04/2025	
Isotope Performance Standard: 13C3-PFBA	6830	ng/L	5000		137	50-200			11/04/2025	
Isotope Dilution Analog: 13C3-PFBS	14.5	ng/L	16.28		89.2	40-135			11/04/2025	
Isotope Dilution Analog: 13C3-PFHxS	14.5	ng/L	16.56		87.6	40-130			11/04/2025	
Isotope Dilution Analog: 13C3-PFPrA	25.2	ng/L	69.85		36.0	20-150			11/04/2025	
Isotope Dilution Analog: 13C4-PFBA	61.1	ng/L	69.85		87.4	10-130			11/04/2025	
Isotope Dilution Analog: 13C4-PFHpA	16.3	ng/L	17.46		93.2	40-130			11/04/2025	
Isotope Performance Standard: 13C4-PFOA	3760	ng/L	2500		150	50-200			11/04/2025	
Isotope Performance Standard: 13C4-PFOS	3440	ng/L	2395		144	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFHxA	15.9	ng/L	17.46		90.8	40-130			11/04/2025	
Isotope Performance Standard: 13C5-PFNA	1910	ng/L	1250		153	50-200			11/04/2025	
Isotope Dilution Analog: 13C5-PFPeA	30.3	ng/L	34.93		86.7	40-130			11/04/2025	
Isotope Dilution Analog: 13C6-PFDA	7.48	ng/L	8.732		85.7	40-130			11/04/2025	
Isotope Dilution Analog: 13C7-PFUnA	7.42	ng/L	8.732		84.9	30-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOA	15.2	ng/L	17.46		86.8	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOS	13.9	ng/L	16.73		83.2	40-130			11/04/2025	
Isotope Dilution Analog: 13C8-PFOSA	18.7	ng/L	17.46		107	40-130			11/04/2025	
Isotope Dilution Analog: 13C9-PFNA	7.33	ng/L	8.732		84.0	40-130			11/04/2025	
Isotope Performance Standard: 18O2-PFHxS	3310	ng/L	2370		140	50-200			11/04/2025	
Isotope Dilution Analog: d3-NMeFOSA	13.5	ng/L	17.46		77.5	10-130			11/04/2025	

Organics-Semivolatiles - Quality Control

Analyte	Result	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Analyzed	Qualifier
---------	--------	-------	-------------	---------------	------	-------------	-----	-----------	----------	-----------

Batch B5J3030 - Method: 1633

Prepared: 10/30/2025

Matrix Spike Dup (B5J3030-MSD1)

Source: 2510354-01

Isotope Dilution Analog: d3-N-MeFOSAA	25.3	ng/L	34.93		72.3	40-170			11/04/2025	
Isotope Dilution Analog: d5-NEtFOSA	10.5	ng/L	17.46		60.2	10-130			11/04/2025	
Isotope Dilution Analog: d5-N-EtFOSAA	34.1	ng/L	34.93		97.8	25-135			11/04/2025	
Isotope Dilution Analog: d7-NMeFOSE	179	ng/L	174.6		103	10-130			11/04/2025	
Isotope Dilution Analog: d9-NEtFOSE	151	ng/L	174.6		86.7	10-130			11/04/2025	



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

Analysis Request - PFAS

For Environmental Water

2810354	Project Name:	Cadillac WWTP - Cadillac PFAS sampling		Matrix:	Water
Location ID:	MI0020257	Activity:	WPFFD	Sample Collector:	Ashley McElmurry
Dept / Division / District:	EGLE-WRD-EPS	Location Code:	C083	Sample Collector Phone:	231-340-0288
State Project Manager:	Grace Borst	Sub Location Code:		Contract Firm Primary Contact:	
State Project Mgr Email:	borstg1@michigan.gov			Primary Contact Phone:	
CC Email 1:	kammers@michigan.gov			Overflow Lab Choice:	
CC Email 2:	mcelmurrya@michigan.gov			*Rush Turnaround Request	
CC Email 3:				Days:	Appr By:

*Project Turnaround Time other than standard (15 business days) must be pre-approved and scheduled with a Laboratory Manager. Details on Page 2.

Lab Use Only	Field Sample ID	Collection Date	Collection Time	Bottle Count	Comments
1	01 C	10/22/25	924	5	
2	02 TB	10/22/25	914	1	
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

ANALYSIS SELECTION

(Applies to all samples listed above)

☒	PFAS - Method 1633 (2x 250 mL plastic bottles per sample with 2 additional bottles per 20 samples for QC)	☐	PFAS - Method 8327 (3x 15 mL plastic test tubes per sample)
-------------------------------------	--	--------------------------	---

Any Known Hazards:

Chain of Custody	Relinquished By	Received By	Date / Time
	Ashley McElmurry EGLE	McElmurry, Ashley	10/23/25 11:20
	Signature	Signature	

ATTACHMENT NAME:

2510354 FINAL EGLE Excel 24 Nov 25 1129.xls

ATTACHMENT TYPE:

Microsoft Office 1-11, Binary format: spreadsheet (XLS)

[illegible]

[illegible]

[illegible]

[illegible]

Collection_Date&Time	SampledBy	SampleReceived	Analysis	Analyte	CAS_Number
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-4:2FTS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-6:2FTS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-8:2FTS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDA	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDaA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDaA+13C2-PFTeDA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFHkA	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFTeDA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C3-HFODA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFBA	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFBS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFHkS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFPtA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFBA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFHpA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFOA	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFOS	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFHkA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFNA	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFPeA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C6-PFDA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C7-PFHhA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOS	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOSA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	13C9-PFNA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	18O2-PFHkS	Isotope Performance Standard
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d3-NMeFOSA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d3-N-MeFOSAA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d5-NEFOSA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d5-N-EtFOSAA	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d7-NMeFOSE	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	d9-NEFOSE	Isotope Dilution Analog
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	11Cl-PF3OUbS	763051-92-9
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	3:3FTCA	356-02-5
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	4:2FTS	757124-72-4
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	5:3FTCA	914637-49-3
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	6:2FTS	27619-97-2
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	7:3FTCA	812-70-4
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	8:2FTS	39108-34-4
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	9Cl-PF3ONS	756426-58-1
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	ADONA	919005-14-4
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	HFPO-DA	13252-13-6
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	NEFOSA	4151-50-2
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	NEFOSAA	2991-50-6
10/22/2025 9:24:00 AM		10/23/2025 11:20:00 AM	PFAS-1633	NEFOSE	1691-99-2

10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NFDHA	151772-58-6
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSA	31506-32-8
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSAA	2355-31-9
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSE	24448-09-7
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBA	375-22-4
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBS	375-73-5
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBSA	30334-69-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDA	335-76-2
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDoA	307-55-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDoS	79780-39-5
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDS	335-77-3
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFED-H	646-83-3
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFEESA	113507-82-7
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFpA	375-85-9
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFpS	375-92-8
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFkA	307-24-4
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFkS	355-46-4
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFkSA	41997-13-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFMBA	863090-89-5
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFMPA	377-73-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFNA	375-95-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFNS	68259-12-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOA	335-67-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOS	1763-23-1
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOSA	754-91-6
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPeA	2706-90-3
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPeS	2706-91-4
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFrA	422-64-0
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFFrS	423-41-6
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFTeDA	376-06-7
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFTrDA	72629-94-8
10/22/2025 9:24:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFLuA	2058-94-8
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-4:2FTS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-6:2FTS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-8:2FTS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDA	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDoA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFDoA+13C2-PFTeDA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFFkA	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C2-PFTeDA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C3-HFPODA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFBA	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFBS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFFkS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C3-PFFrA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFBA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFFpA	Isotope Dilution Analog

10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFOA	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C4-PFOS	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFHxA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFNA	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C5-PFPeA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C6-PFDA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C7-PFHxA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOS	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C8-PFOSA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	13C9-PFNA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	18O2-PFHxS	Isotope Performance Standard
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d3-NMeFOSA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d3-N-MeFOSAA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d5-NMeFOSA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d5-N-MeFOSAA	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d7-NMeFOSE	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	d9-NMeFOSE	Isotope Dilution Analog
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	11Cl-PF3OUtS	763051-92-9
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	3:3FTCA	356-02-5
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	4:2FTS	757124-72-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	5:3FTCA	914637-49-3
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	6:2FTS	27619-97-2
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	7:3FTCA	812-70-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	8:2FTS	39108-34-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	9Cl-PF3ONS	756426-58-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	ADONA	919005-14-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	HFPO-DA	13252-13-6
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSA	4151-50-2
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSAA	2991-50-6
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSE	1691-99-2
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NFDHA	151772-58-6
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSA	31506-32-8
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSAA	2355-31-9
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	NMeFOSE	24448-09-7
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBA	375-22-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBS	375-73-5
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFBSA	30334-69-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDA	335-76-2
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDoA	307-55-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDoS	79780-39-5
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFDS	335-77-3
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFECHS	646-83-3
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFEESA	113507-82-7
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFHpA	375-85-9
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFHpS	375-92-8
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFHxA	307-24-4

10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfHxS	355-46-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfHxSA	41997-13-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfMBA	863090-89-5
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfMPA	377-73-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfNA	375-95-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PfNS	68259-12-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOA	335-67-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOS	1763-23-1
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFOSA	754-91-6
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPeA	2706-90-3
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPeS	2706-91-4
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPtA	422-64-0
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFPtS	423-41-6
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFTeDA	376-06-7
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFTtDA	72629-94-8
10/22/2025 9:14:00 AM	10/23/2025 11:20:00 AM	PFAS-1633	PFluA	2058-94-8

[illegible]

[illegible]

[illegible]

Qualifier_Text

Result is estimated/suspect due to isotope dilution analog recovery in the sample outside of QC limits.

Result is estimated/suspect due to isotope dilution analog recovery in the sample outside of QC limits.

Result(s) and reporting limit(s) are estimated due to low matrix spike recovery.

